

HIGHLIGHTS OF PRESCRIBING INFORMATION

These highlights do not include all the information needed to use FERRIPROX 1000 mg delayed-release tablets safely and effectively. See full prescribing information for FERRIPROX delayed-release tablets. FERRIPROX® 1000 mg delayed-release tablets (deferiprone), for oral use

WARNING: AGRANULOCYTOSIS AND NEUTROPENIA

See full prescribing information for complete boxed warning.

- FERRIPROX can cause agranulocytosis that can lead to serious infections and death. Neutropenia may precede the development of agranulocytosis. (5.1)
- Measure the absolute neutrophil count (ANC) before starting FERRIPROX and monitor regularly while on therapy. (5.1)
- Interrupt FERRIPROX if neutropenia develops. (5.1)
- Interrupt FERRIPROX if infection develops and monitor the ANC more frequently. (5.1)
- Advise patients taking FERRIPROX to report immediately any symptoms indicative of infection. (5.1)

INDICATIONS AND USAGE

FERRIPROX is indicated for the treatment of iron overload in patients with thalassemia major when deferoxamine therapy is contra-indicated or inadequate.

DOSAGE AND ADMINISTRATION

- Starting oral dosage: 75 mg/kg/day (actual body weight) in two divided doses (2.2)
- Maximum oral dosage: 99 mg/kg/day (actual body weight) in two divided doses (2.2)

DOSAGE FORMS AND STRENGTHS

- Delayed-release tablets: 1000 mg with functional scoring (3)

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CONTRAINDICATIONS

Hypersensitivity to deferiprone or to any of the excipients in the formulations. (4)

WARNINGS AND PRECAUTIONS

- Liver Enzyme Elevations: Monitor monthly and discontinue for persistent elevations. (5.2)
- Zinc Deficiency: Monitor during therapy and supplement for deficiency. (5.3)
- Embryo-Fetal Toxicity: Can cause fetal harm. (5.4)

ADVERSE REACTIONS

- The most common adverse reactions in patients with thalassemia (incidence $\geq 6\%$) are nausea, vomiting, abdominal pain, arthralgia, ALT increased and neutropenia. (5.1, 6)

DRUG INTERACTIONS

- Drugs Associated with Neutropenia or Agranulocytosis: Avoid co-administration. If co-administration is unavoidable, closely monitor the absolute neutrophil count. (7.1)
- UGT1A6 Inhibitors: Avoid co-administration. (7.2)
- Polyvalent Cations: Allow at least a 4-hour interval between administration of FERRIPROX and drugs or supplements containing polyvalent cations (e.g., iron, aluminum, or zinc). (2.2, 7.2)

USE IN SPECIFIC POPULATIONS

Lactation: Advise not to breastfeed. (8.2)

Revised: 05/2023

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FULL PRESCRIBING INFORMATION

WARNING: AGRANULOCYTOSIS AND NEUTROPENIA

- FERRIPROX can cause agranulocytosis that can lead to serious infections and death. Neutropenia may precede the development of agranulocytosis. *[see Warnings and Precautions (5.1)]*
- Measure the absolute neutrophil count (ANC) before starting FERRIPROX therapy and monitor regularly while on therapy.
- Interrupt FERRIPROX therapy if neutropenia develops. *[see Warnings and Precautions (5.1)]*
- Interrupt FERRIPROX if infection develops, and monitor the ANC more frequently. *[see Warnings and Precautions (5.1)]*
- Advise patients taking FERRIPROX to report immediately any symptoms indicative of infection. *[see Warnings and Precautions (5.1)]*

1 INDICATIONS AND USAGE

FERRIPROX is indicated for the treatment of iron overload in patients with thalassemia major when deferoxamine therapy is contraindicated or inadequate.

2 DOSAGE AND ADMINISTRATION

2.1 Monitoring for Safety

Due to the risk of agranulocytosis, monitor ANC before and during FERRIPROX therapy.

Test ANC prior to start of FERRIPROX therapy and monitor on the following schedule during treatment:

- First six months of therapy: Monitor ANC weekly;
- Next six months of therapy: Monitor ANC once every two weeks;
- After one year of therapy: Monitor ANC every two to four weeks (or at the patient's blood transfusion interval in patients that have not experienced an interruption due to any decrease in ANC *[see Warnings and Precautions (5.1)]*).

Due to the risk of hepatic transaminase elevations, monitor ALT before and monthly during FERRIPROX therapy *[see Warnings and Precautions (5.2)]*.

Due to the risk of zinc deficiency, monitor zinc levels before and regularly during FERRIPROX therapy *[see Warnings and Precautions (5.3)]*.

2.2 Recommended Dosage

Starting Dosage

The recommended starting oral dosage of FERRIPROX delayed-release tablets is 75 mg/kg/day (actual body weight) in two divided doses per day (taken approximately 12 hours apart), with food. Round the total daily dose to the nearest 500 mg (half-tablet). Table 1 describes the number of FERRIPROX delayed-release tablets needed to achieve the 75 mg/kg/day total starting daily dosage.

Table 1: Number of FERRIPROX delayed-release tablets Needed to Achieve the Total Starting Daily Dosage of 75 mg/kg (rounded to the nearest half-tablet)		
Body Weight (kg)	Morning	Evening
20	0.5	1
30	1	1.5
40	1.5	1.5
50	2	2
60	2	2.5
70	2.5	3
80	3	3
90	3.5	3.5

To minimize gastrointestinal upset when first starting therapy, dosing can start at 45 mg/kg/day and increase weekly by 15 mg/kg/day increments until the full prescribed dose is achieved.

For patients who have trouble swallowing tablets, consider the use of FERRIPROX Oral Solution (see the prescribing information for FERRIPROX Oral Solution).

Dosage Adjustments

Tailor dosage adjustments of FERRIPROX delayed-release tablets to the individual patient's response and therapeutic goals (maintenance or reduction of body iron burden). The maximum total daily oral dosage is 99 mg/kg (actual body weight) divided into two doses taken approximately 12 hours apart with food. Table 2 describes the number of FERRIPROX delayed-release tablets needed to achieve the 99 mg/day total maximum daily dosage.

Table 2: Number of FERRIPROX delayed-release tablets Needed to Achieve a Total Maximum Recommended Daily Dosage of 99 mg/kg (rounded to the nearest half-tablet)		
Body Weight (kg)	Morning	Evening
20	1	1
30	1.5	1.5
40	2	2
50	2.5	2.5
60	3	3
70	3.5	3.5
80	4	4
90	4.5	4.5

2.3 Monitoring Ferritin Levels to Assess Efficacy

Monitor serum ferritin concentration every two to three months to assess the effect of FERRIPROX on body iron stores. If the serum ferritin is consistently below 500 mcg/L, consider temporarily interrupting FERRIPROX therapy until serum ferritin rises above 500 mcg/L.

2.4 Dosage Modification for Drug Interactions

Allow at least a 4-hour interval between administration of FERRIPROX and other drugs or supplements containing polyvalent cations such as iron, aluminum, or zinc [see *Drug Interactions (7.2)*, *Clinical Pharmacology (12.3)*].

3 DOSAGE FORMS AND STRENGTHS

- delayed-release tablets: 1,000 mg, capsule-shaped, white to off-white tablets with functional scoring, engraved “FPX” bisect “DR” on one side, “APO” bisect “1000” on the other”.

4 CONTRAINDICATIONS

FERRIPROX is contraindicated in patients with known hypersensitivity to deferiprone or to any of the excipients in the formulations. The following reactions have been reported in association with the administration of deferiprone: Henoch-Schönlein purpura; urticaria; and periorbital edema with skin rash [see *Adverse Reactions (6.2)*].

5 WARNINGS AND PRECAUTIONS

5.1 Agranulocytosis and Neutropenia

Fatal agranulocytosis can occur with FERRIPROX use. FERRIPROX can also cause neutropenia, which may foreshadow agranulocytosis. Measure the absolute neutrophil count (ANC) before starting FERRIPROX therapy and monitor it regularly while on therapy [see *Dosage and Administration (2.1)*].

Reduction in the frequency of ANC monitoring should be considered on an individual patient basis, according to the health care provider's assessment of the patient's understanding of the risk minimization measures required during therapy.

Interrupt FERRIPROX therapy if neutropenia develops ($ANC < 1.5 \times 10^9/L$).

Interrupt FERRIPROX if infection develops and monitor the ANC frequently.

Advise patients taking FERRIPROX to immediately interrupt therapy and report to their physician if they experience any symptoms indicative of infection.

The incidence of agranulocytosis was 1.7% of patients in pooled clinical trials of 642 patients with thalassemia syndromes. The mechanism of FERRIPROX-associated agranulocytosis is unknown. Agranulocytosis and neutropenia usually resolve upon discontinuation of FERRIPROX, but there have been reports of agranulocytosis leading to death.

Implement a plan to monitor for and to manage agranulocytosis and neutropenia prior to initiating FERRIPROX treatment.

For agranulocytosis ($ANC < 0.5 \times 10^9/L$):

Consider hospitalization and other management as clinically appropriate.

Do not resume FERRIPROX in patients who have developed agranulocytosis unless potential benefits outweigh potential risks. Do not rechallenge patients who have developed neutropenia with FERRIPROX unless potential benefits outweigh potential risks.

For neutropenia ($ANC < 1.5 \times 10^9/L$ and $> 0.5 \times 10^9/L$):

Instruct the patient to immediately discontinue FERRIPROX and all other medications with a potential to cause neutropenia.

Obtain a complete blood cell (CBC) count, including a white blood cell (WBC) count corrected for the presence of nucleated red blood cells, an absolute neutrophil count (ANC), and a platelet count daily until recovery ($ANC \geq 1.5 \times 10^9/L$).

5.2 Liver Enzyme Elevations

In pooled clinical trials, 7.5% of 642 patients with thalassemia syndromes treated with FERRIPROX developed increased ALT values. Four (0.62%) FERRIPROX-treated subjects discontinued the drug due to increased serum ALT levels and 1 (0.16%) due to an increase in both ALT and AST.

Monitor serum ALT values monthly during therapy with FERRIPROX and consider interruption of therapy if there is a persistent increase in the serum transaminase levels [see *Dosage and Administration* (2.1)].

5.3 Zinc Deficiency

Decreased plasma zinc concentrations have been observed on FERRIPROX therapy. Monitor plasma zinc annually, and supplement in the event of a deficiency [see *Dosage and Administration* (2.1)].

5.4 Embryo-Fetal Toxicity

Based on findings from animal reproduction studies and evidence of genotoxicity, FERRIPROX can cause fetal harm when administered to a pregnant woman. The available data on the use of FERRIPROX in pregnant women are insufficient to inform risk. In animal studies, administration of deferiprone during the period of organogenesis resulted in embryo-fetal death and malformations at doses lower than equivalent human clinical doses. Advise pregnant women and females of reproductive potential of the potential risk to the fetus [see *Use in Specific Populations* (8.1)].

Advise females of reproductive potential to use an effective method of contraception during treatment with FERRIPROX and for at least six months after the last dose. Advise males with female partners of reproductive potential to use effective contraception during treatment with FERRIPROX and for at least three months after the last dose [see *Use in Specific Populations* (8.1, 8.3)].

6 ADVERSE REACTIONS

The following clinically significant adverse reactions are described below and elsewhere in the labeling:

- Agranulocytosis and Neutropenia [see *Warnings and Precautions* (5.1)]
- Liver Enzyme Elevations [see *Warnings and Precautions* (5.2)]
- Zinc Deficiency [see *Warnings and Precautions* (5.3)]

6.1 Clinical Trial Experience

Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in practice.

FERRIPROX delayed-release tablets were evaluated in trials in healthy subjects. FERRIPROX delayed-release tablets contain deferiprone, the same active ingredient as FERRIPROX film-coated tablets (deferiprone) and FERRIPROX oral solution (deferiprone).

The following adverse reaction information represents the pooled data collected from single arm or active-controlled clinical trials with FERRIPROX film-coated tablets (deferiprone) or FERRIPROX oral Solution (deferiprone).

The safety of FERRIPROX was evaluated in the pooled clinical trial database [see *Clinical Studies* (14.1)]. Patients received FERRIPROX film-coated tablets or FERRIPROX oral Solution. FERRIPROX was administered orally three times a day (total daily

dose either 50, 75, or 99 mg/kg), N=642. Among 642 patients receiving FERRIPROX, 492 (76.6%) were exposed for 6 months or longer and 365 (56.9%) were exposed for greater than one year.

The median age of patients who received FERRIPROX was 19 years (range 1, 77 years); 50.2% female; 71.2% White, 17.8% Asian, 9.2% Unknown, 1.2% Multi-racial and 0.6% Black.

The most serious adverse reaction reported in clinical trials with FERRIPROX was agranulocytosis [see *Warnings and Precautions (5.1)*].

The most common adverse reactions ($\geq 6\%$) reported during clinical trials were nausea, vomiting, abdominal pain, arthralgia, alanine aminotransferase increased and neutropenia.

The table below lists the adverse drug reactions that occurred in at least 1% of patients treated with FERRIPROX in clinical trials in patients with thalassemia syndromes.

Table 3: Adverse reactions occurring in $\geq 1\%$ of FERRIPROX-treated patients with thalassemia syndromes

Body System Adverse Reaction	(N=642) % Patients
BLOOD AND LYMPHATIC SYSTEM DISORDERS	
Neutropenia	6
Agranulocytosis	2
GASTROINTESTINAL DISORDERS	
Nausea	13
Abdominal pain/discomfort	10
Vomiting	10
Diarrhea	3
Dyspepsia	2
INVESTIGATIONS	
Alanine aminotransferase increased	7
Weight increased	2
Aspartate aminotransferase increased	1
METABOLISM AND NUTRITION DISORDERS	
Increased appetite	4
Decreased appetite	1
MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS	
Arthralgia	10
Back pain	2
Pain in extremity	2
Arthropathy	1
NERVOUS SYSTEM DISORDERS	
Headache	2

Gastrointestinal symptoms such as nausea, vomiting, and abdominal pain were the most frequent adverse reactions reported by patients participating in clinical trials and led to the discontinuation of FERRIPROX therapy in 1.6% of patients.

Chromaturia (reddish/brown discoloration of the urine) is a result of the excretion of iron in the urine.

6.2 Postmarketing Experience

The following additional adverse reactions have been reported in patients receiving FERRIPROX. Because these reactions are reported voluntarily from a population of uncertain size, it is not always possible to reliably estimate their frequency or to establish a causal relationship to drug exposure.

Blood and lymphatic system disorders: thrombocytosis, pancytopenia.

Cardiac disorders: atrial fibrillation, cardiac failure.

Congenital, familial and genetic disorders: hypospadias.

Eye disorders: diplopia, papilledema, retinal toxicity.

Gastrointestinal disorders: enterocolitis, rectal hemorrhage, gastric ulcer, pancreatitis, parotid gland enlargement.

General disorders and administration site conditions: chills, edema peripheral, multi-organ failure.

Hepatobiliary disorders: jaundice, hepatomegaly.

Immune system disorders: anaphylactic shock, hypersensitivity.

Infections and infestations: cryptococcal cutaneous infection, enteroviral encephalitis, pharyngitis, pneumonia, sepsis, furuncle, infectious hepatitis, rash pustular, subcutaneous abscess.

Investigations: blood bilirubin increased, blood creatinine phosphokinase increased.

Metabolism and nutrition disorders: metabolic acidosis, dehydration.

Musculoskeletal and connective tissue disorders: myositis, chondropathy, trismus.

Nervous system disorders: cerebellar syndrome, cerebral hemorrhage, convulsion, gait disturbance, intracranial pressure increased, psychomotor skills impaired, pyramidal tract syndrome, somnolence.

Psychiatric disorders: bruxism, depression, obsessive-compulsive disorder.

Renal disorders: glycosuria, hemoglobinuria.

Respiratory, thoracic and mediastinal disorders: acute respiratory distress syndrome, epistaxis, hemoptysis, pulmonary embolism.

Skin, subcutaneous tissue disorders: hyperhidrosis, periorbital edema, photosensitivity reaction, pruritis, urticaria, rash, Henoch-Schönlein purpura.

Vascular disorders: hypotension, hypertension.

7 DRUG INTERACTIONS

7.1 Drugs Associated with Neutropenia or Agranulocytosis

Avoid co-administration of FERRIPROX with other drugs known to be associated with neutropenia or agranulocytosis. If co-administration is unavoidable, closely monitor the absolute neutrophil count [see *Warnings and Precautions* (5.1)].

7.2 Effect of Other Drugs on FERRIPROX

UDP-Glucuronosyltransferases (UGT)

Avoid use of UGT1A6 inhibitors (e.g., diclofenac, probenecid, or silymarin (milk thistle)) with FERRIPROX [see *Dosage and Administration* (2.2), *Adverse Reactions* (6.1), *Clinical Pharmacology* (12.3)].

Polyvalent Cations

Deferiprone has the potential to bind polyvalent cations (e.g., iron, aluminum, and zinc); allow at least a 4-hour interval between FERRIPROX and other medications (e.g., antacids), or supplements containing these polyvalent cations [see *Dosage and Administration* (2.2)].

8 USE IN SPECIFIC POPULATIONS

8.1 Pregnancy

Risk Summary

In animal reproduction studies, oral administration of deferiprone to pregnant rats and rabbits during organogenesis at doses 33% and 49%, respectively, of the maximum recommended human dose (MRHD) resulted in structural abnormalities, embryo-fetal mortality and alterations to growth (*see Data*). The limited available data from deferiprone use in pregnant women are insufficient to inform a drug-associated risk of major birth defects and miscarriage. Based on evidence and developmental toxicity in animal studies, FERRIPROX can cause fetal harm when administered to a pregnant woman. Advise pregnant women and females of reproductive potential of the potential risk to a fetus.

The estimated background risk of major birth defects and miscarriage for the indicated population is unknown. All pregnancies have a background risk of birth defect, loss, or other adverse outcomes. In the U.S. general population, the estimated background risk of major birth defects and of miscarriage is 2-4% and 15-20%, respectively.

Data

Human Data

Post-marketing data available from 39 pregnancies of deferiprone-treated patients and 10 pregnancies of partners of deferiprone-treated patients are as follows:

Of the 39 pregnancies in deferiprone-treated patients, 23 resulted in healthy newborns, 6 ended in spontaneous abortion, 9 had unknown outcomes, and 1 infant was born with anal atresia, nephroptosis, ventricular septal defect, hemivertebra and urethral fistula.

Of the 10 pregnancies in partners of deferiprone-treated patients, 5 resulted in healthy newborns, 1 resulted in a healthy newborn with slight hypospadias, 1 was electively terminated, 1 resulted in the intrauterine death of twins, and 2 had unknown outcomes.

Animal Data

During organogenesis, pregnant rats and rabbits received deferiprone at oral doses of 0, 30, 80 or 200 mg/kg/day, and 0, 10, 50, or 150 mg/kg/day, respectively. The daily dose was administered as two equal divided doses approximately 7 hours apart. Doses of 200 mg/kg/day in rats and 150 mg/kg/day in rabbits, approximately 33% and 49% of the MRHD, respectively, resulted in increased post-implantation loss and reduced fetal weights in the presence of maternal toxicity (reduced maternal body weight and body weight gain in both rats and rabbits; abnormal large placenta at low incidence in rats). The 200 mg/kg/day dose in rats resulted in external, visceral and skeletal fetal malformations such as cranial malformations, cleft palate, limb malrotation, anal atresia, internal hydrocephaly, anophthalmia and fused bones. The dose of 150 mg/kg/day in rabbits resulted in external fetal malformations (partially opened eyes) and minor blood vessel and skeletal variations.

In rats, malformations including micrognathia and persistent ductus arteriosus could be observed in the absence of maternal toxicity at doses equal to or greater than 30 and 80 mg/kg/day, approximately 5% and 13% of the MHRD, respectively.

8.2 Lactation

Risk Summary

There is no information regarding the presence of deferiprone in human milk, the effects on the breastfed child, or the effects on milk production.

Because of the potential for serious adverse reactions in the breastfed child, including the potential for tumorigenicity shown for deferiprone in animal studies, advise patients that breastfeeding is not recommended during treatment with FERRIPROX, and for at least 2 weeks after the last dose.

8.3 Females and Males of Reproductive Potential

Pregnancy Testing

Pregnancy testing is recommended for females of reproductive potential prior to initiating FERRIPROX.

Contraception

Females

FERRIPROX can cause embryo-fetal harm when administered to a pregnant woman [see *Use in Specific Populations (8.1)*]. Advise female patients of reproductive potential to use effective contraception during treatment with FERRIPROX and for at least 6 months after the last dose.

Males

Based on genotoxicity findings, advise males with female partners of reproductive potential to use effective contraception during treatment with FERRIPROX and for at least 3 months after the last dose [see *Nonclinical Toxicology (13.1)*].

8.4 Pediatric Use

Limited information is available on the use of deferiprone in children between 6 and 10 years of age, and very limited information is available on deferiprone use in children under 6 years of age.

8.5 Geriatric Use

Clinical studies of deferiprone did not include sufficient numbers of subjects aged 65 and over to determine whether they respond differently from younger subjects. Other reported clinical experience has not identified differences in responses between the elderly and younger patients.

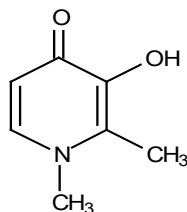
10 OVERDOSAGE

No cases of acute overdose have been reported. There is no specific antidote to FERRIPROX overdose.

Neurological disorders such as cerebellar symptoms, diplopia, lateral nystagmus, psychomotor slowdown, hand movements and axial hypotonia have been observed in children treated with 2.5 to 3 times the recommended dose for more than one year. The neurological disorders progressively regressed after deferiprone discontinuation.

11 DESCRIPTION

FERRIPROX delayed-release tablets contain 1,000 mg deferiprone (3-hydroxy-1,2-dimethylpyridin-4-one), a synthetic, orally active, iron-chelating agent. The molecular formula for deferiprone is $C_7H_9NO_2$ and its molecular weight is 139.15 g/mol. Deferiprone has the following structural formula:



Deferiprone is a white to pinkish-white powder. It is sparingly soluble in deionized water (14.3 mg/mL) and has a melting point range of 272 °C - 278 °C.

White to off-white, capsule-shaped tablets, and imprinted with “FPX” score “DR” on one side and “APO” score “1000” on the other. The tablets can be broken in half along the score line. Each tablet contains 1,000 mg deferiprone and the following inactive ingredients: Tablet core - hypromellose acetate succinate, magnesium oxide, colloidal silicon dioxide and magnesium stearate; Coating - triethyl citrate, talc, titanium dioxide, and methacrylic acid and ethyl acrylate copolymer.

12 CLINICAL PHARMACOLOGY

12.1 Mechanism of Action

Deferiprone is a chelating agent with an affinity for ferric ions (iron III). Deferiprone binds with ferric ions to form neutral 3:1 (deferiprone:iron) complexes that are stable at physiological pH.

12.2 Pharmacodynamics

No clinical studies were performed to assess the relationship between the dose of deferiprone and the amount of iron eliminated from the body.

Cardiac Electrophysiology

At the maximum approved recommended dose, deferiprone does not prolong the QT interval to any clinically relevant extent.

12.3 Pharmacokinetics

In healthy subjects, the mean $\pm$ SD C_{max} of deferiprone in serum was 6 ± 2 mcg/mL, and the mean $\pm$ SD AUC was 28 ± 7 mcg·h/mL following oral administration of a 1,000 mg dose of FERRIPROX delayed-release tablets with food.

Absorption

Peak serum concentrations of deferiprone occur approximately 2 hours after a single dose of FERRIPROX delayed-release tablets in fasted healthy subjects.

Effect of Food

Following the administration of FERRIPROX delayed-release tablets to healthy volunteers, the C_{max} and the AUC of deferiprone remain unchanged after a high-fat meal (approximately 1,000 calories, 53% fat, 33% carbohydrates, and 14% protein) compared to fasted conditions.

Effect of Alcohol

At 40% (v/v) alcohol concentration *in vitro* dissolution studies, there was 88% release of deferiprone from a 1,000 mg FERRIPROX delayed-release tablet within two hours compared to 4% release of deferiprone within 2 hours in the absence of alcohol.

Distribution

The apparent mean $\pm$ SD volume of distribution (V/F) of deferiprone was 97 ± 28 L following oral administration of a 1,000 mg dose of FERRIPROX delayed-release tablets with food.

Elimination

The mean $\pm$ SD elimination half-life of deferiprone is 1.8 ± 0.3 hours following the administration of FERRIPROX delayed-release tablets.

Metabolism

Deferiprone is metabolized primarily by UGT1A6. The major metabolite of deferiprone is the 3-*O*-glucuronide, which lacks iron-binding capability.

Excretion

Following oral administration, 75% to 90% of the administered dose was recovered in urine (primarily as metabolite) in the first 24 hours.

Specific Populations

No clinically significant differences in the pharmacokinetics of deferiprone were observed based on sex, race/ethnicity, body weight, mild to severe (eGFR 15 to 89 mL/min/1.73 m²) renal impairment, or mild (Child Pugh Class A) to moderate (Child Pugh Class B) hepatic impairment. The effect of age, including geriatric or pediatric populations, end stage renal disease or severe (Child Pugh Class C) hepatic impairment on the pharmacokinetics of deferiprone is unknown.

Drug Interaction Studies

In Vitro Studies

UGT1A6 Inhibitors: Phenylbutazone (UGT1A6 inhibitor) decreased glucuronidation of deferiprone by up to 78%.

Polyvalent Cations: Deferiprone has the potential to bind polyvalent cations (e.g., iron, aluminum, and zinc).

13 NONCLINICAL TOXICOLOGY

13.1 Carcinogenesis, Mutagenesis, Impairment of Fertility

Carcinogenicity studies have not been conducted with deferiprone. However, in view of the genotoxicity results, and the findings of mammary gland hyperplasia and mammary gland tumors in rats treated with deferiprone in the 52-week toxicology study, tumor formation in carcinogenicity studies must be regarded as likely.

Deferiprone was positive in a mouse lymphoma cell assay *in vitro*. Deferiprone was clastogenic in an *in vitro* chromosomal aberration test in mice and in a chromosomal aberration test in Chinese Hamster Ovary cells. Deferiprone given orally or intraperitoneally was clastogenic in a bone marrow micronucleus assay in non-iron-loaded mice. A micronucleus test was also positive when mice predosed with iron dextran were treated with deferiprone. Deferiprone was not mutagenic in the Ames bacterial reverse mutation test.

A fertility and early embryonic development study of deferiprone was conducted in rats. Sperm counts, motility and morphology were unaffected by treatment with deferiprone. There were no effects observed on male or female fertility or reproductive function at the highest dose which was 25% of the MRHD.

14 CLINICAL STUDIES

FERRIPROX delayed-release tablets were evaluated in trials in healthy subjects. FERRIPROX delayed-release tablets contain deferiprone, the same active ingredient as FERRIPROX film-coated tablets and FERRIPROX oral solution. The following information is based on studies with FERRIPROX film-coated tablets (deferiprone) (three times a day) and FERRIPROX oral solution (deferiprone).

In a prospective, planned, pooled analysis of patients with thalassemia syndromes from several studies, the efficacy of deferiprone was assessed in transfusion-dependent iron overload patients in whom previous iron chelation therapy had failed or was considered inadequate due to poor tolerance. The main criterion for chelation failure was serum ferritin > 2,500 mcg/L before treatment with deferiprone. Deferiprone therapy (35-99 mg/kg/day) was considered successful in individual patients who experienced a ≥ 20% decline in serum ferritin within one year of starting therapy.

Data from a total of 236 patients were analyzed. Of the 224 patients with thalassemia who received deferiprone monotherapy and were eligible for serum ferritin analysis, 105 (47%) were male and 119 (53%) were female. The mean age of these patients was 18.2 years (range 2 to 62; 91 patients were <17).

For the patients in the analysis, the endpoint of at least a 20% reduction in serum ferritin was met in 50% (of 236 subjects), with a 95% confidence interval of 43% to 57%.

A small number of patients with thalassemia and iron overload were assessed by measuring the change in the number of milliseconds (ms) in the cardiac MRI T2* value before and after treatment with deferiprone for one year. There was an increase in cardiac MRI T2* from a mean at baseline of 11.8 ± 4.9 ms to a mean of 15.1 ± 7.0 ms after approximately one year of treatment. The clinical significance of this observation is not known.

16 HOW SUPPLIED/STORAGE AND HANDLING

FERRIPROX® (deferiprone) delayed-release tablets are white to off-white capsule-shaped, beveled edge, biconvex coated tablets, and have a functional score engraved “FPX” bisect “DR” on one side, “APO” bisect “1000” on the other. They are supplied in child-resistant blister packs or HDPE bottles.

Do not store above 30°C.

Keep out of the reach and sight of children.

Product Registration Holder, Importer & Distributor:

Pharmaforte (M) Sdn Bhd, 2, Jalan PJU 3/49, Sunway Damansara, 47810 Petaling Jaya, Selangor, Malaysia.

Manufactured by:

Apotex Inc. – Etobicoke Site, Canada

<Chiesi logo>

Product Owner:

Chiesi Farmaceutici S.p.A., Via Palermo 26/A, 43122 Parma, Italy